

DIC

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Disseminated Intravascular Coagulation (DIC)

Dr Onuoha – Paediatrics, topic 129

Outline

Introduction · Aetiology (triggers) · Normal clotting vs what goes wrong · Pathophysiology (two phases) · Clinical features · Investigations · Complications · Treatment · Prognosis

Introduction

DIC is a **fatal condition that results in abnormal clotting of blood in all parts of the body**. It is **life-threatening if left untreated**.

Aetiology

DIC is not a disease in itself – it is a **final common pathway** that occurs when the body's clotting system goes out of control. Think of it as: the body **presses the “clotting button” everywhere at once**, then **runs out of clotting materials**, and then **starts bleeding uncontrollably**.

DIC always starts from **something big that shocks the body** or conditions that abnormally activate clotting:

1. **Severe infection (sepsis)** – bacteria release toxins -> massive inflammation.
2. **Trauma / injury** – tissue damage releases clotting activators.
3. **Malignancy** – cancer cells can activate clotting pathways.

4. **Envenomation** – some venoms directly activate clotting.
5. **Medical complications** – transfusions, surgeries, obstetric complications.

Normal clotting vs what goes wrong

Normally clotting is **controlled and limited**: injury occurs -> clotting is **localised to the injury site** -> platelets and clotting factors form a clot -> bleeding stops -> the system shuts off.

In **DIC** this control is lost and the process becomes **generalised**.

Pathophysiology – two phases

Phase 1 – excessive clotting. A trigger (e.g. sepsis) releases **pro-coagulant substances** -> the **entire clotting cascade is activated throughout the body** -> **tiny clots form everywhere, especially in small vessels**. This causes **vaso-occlusion** (blocked vessels), **tissue ischaemia**, and damage to organs – especially the **kidneys, brain and lungs**.

Critical point – consumption. While forming all these clots, **platelets and clotting factors are used up**. This is called **consumption coagulopathy**.

Phase 2 – bleeding. Now the body has **no platelets and no clotting factors**, so **even small injuries cause massive bleeding** – the patient bleeds from orifices.

Simple flow: trigger -> mass clotting -> consumption of platelets/factors -> no clotting ability -> severe bleeding.

Why both clotting AND bleeding happen (the key confusion point): **inside the vessels there is clotting; outside/at wounds there is bleeding** – because all the clotting resources have already been used up internally.

Clinical features

1. **Bleeding from orifices** – from gums, IV sites, wounds and body orifices.
2. **Clotting effects** – organ damage due to blocked vessels.
3. **Systemic effects** – **hypotension** (from blood loss), **shock** (from bleeding and poor organ perfusion), and **coma** (from brain involvement).

Investigations

- **Platelets down** – used up in clotting.
- **Clotting factors down** – prolonged **PT and aPTT**.
- **D-dimer up** – indicates clot breakdown.
- **FBC** – may show **anaemia** (from bleeding).

Complications

- **From clotting:** acute kidney injury, organ failure, cardiorespiratory failure.
- **From bleeding:** severe anaemia, shock, coma and death.
- **Combined:** multi-organ failure.

Treatment

Treat DIC in **two directions**:

1. **Replace what is lost** – platelets, fresh frozen plasma (clotting factors), and blood transfusion.
2. **Stop the cause (most important)** – treat sepsis with antibiotics, remove dead tissue, treat the malignancy, and consider **plasmapheresis** (removes harmful substances by removing the plasma content of the patient's blood). **If you don't stop the cause, DIC continues.**

Prognosis

Prognosis is **generally poor**. Mortality is high because of the **rapid progression** of DIC, the **difficulty of replacing clotting factors and platelets quickly** (availability challenges), and the **severity of the underlying disease** (e.g. septic shock). DIC is a **common pathway to death** from septic shock and other underlying causes.

References

Lecture material of Dr Onuoha (500-level Paediatrics, Abia State University); Nelson Textbook of Paediatrics.